

Discounts cost store CA_1 \$69.6K in profit

5 years of real Walmart sales (M5 dataset, Jan 2011 to May 2016). Food shoppers barely react to price, so 14,016 discount weeks gave away margin, while SNAP benefit weeks lifted sales in the two biggest food departments by 13.5–18.6% for free.

–\$69.6K

Profit lost to discounts at a 35% margin. Ranges from –\$73.8K at 25% to –\$65.4K at 45%.

\$22.9M

Revenue analysed

3,049

Products, 7 departments

14,016

Discount Weeks tested

-0.59

Food Elasticity elasticity

+13.5–18.6%

SNAP week lift, FOODS_2 & 3

The Problem

- Discounts are run without knowing if they pay back.
- Lower prices may only subsidise sales that would happen anyway.
- Promotion timing ignores when customers already have money to spend.
- No evidence per department on price sensitivity.

Our Approach

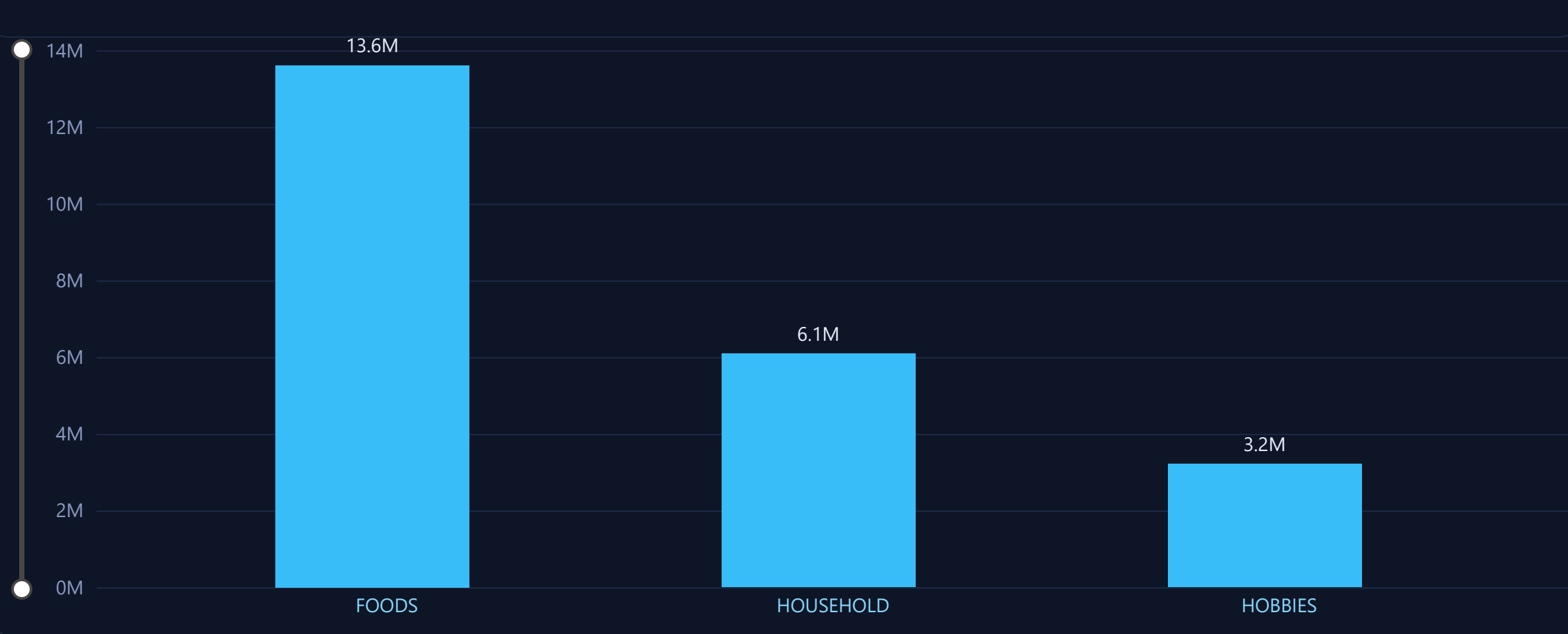
- Measures price elasticity per department from real price changes.
- Controls for SNAP days, events, season and trend.
- Validates on 52 weeks the model never saw.
- Prices every real discount week at 25%, 35% and 45% margins.

The Result

- Food elasticity is only **–0.59**: a 10% discount adds about 6% units.
- Discounts lost **\$65K to \$74K** profit at every margin tested.
- SNAP weeks lift FOODS_2 and FOODS_3 sales 13.5–18.6% with no discount.
- Clear rules: Stop routine food discounts and avoid SNAP weeks.

Revenue by category

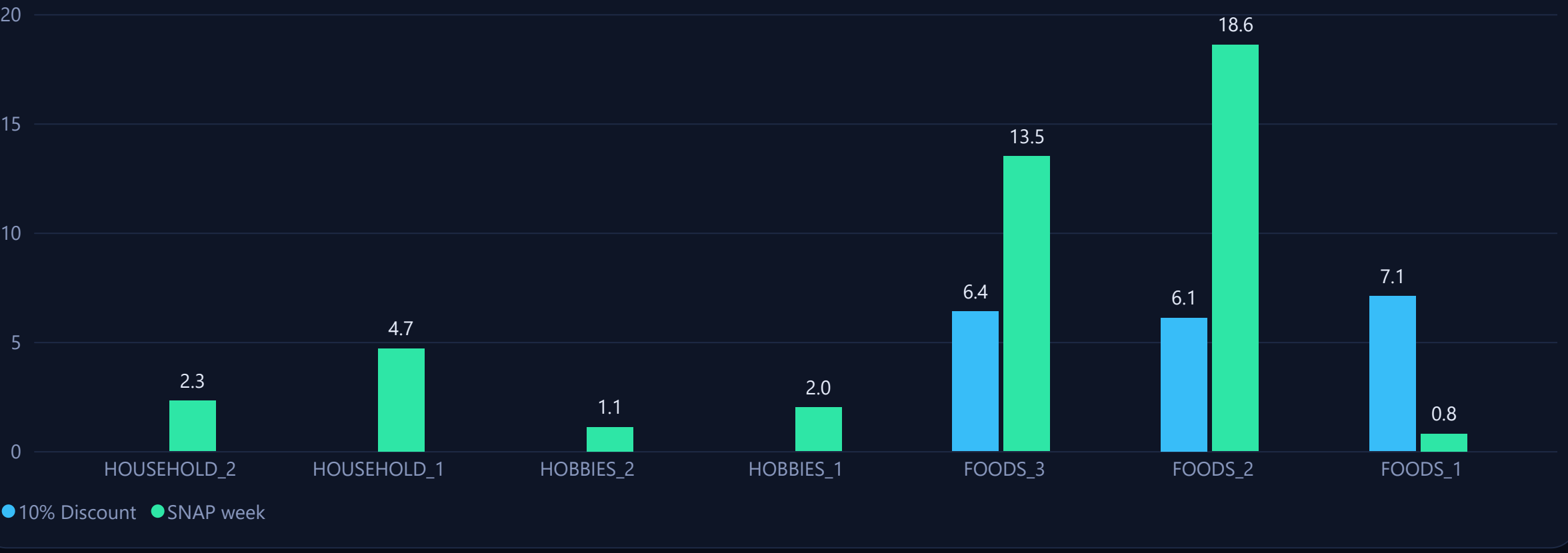
Store CA_1, Jan 2011 to May 2016 (US \$). Food is 59% of sales, so food discount decisions matter most.



A 10% discount vs a SNAP week

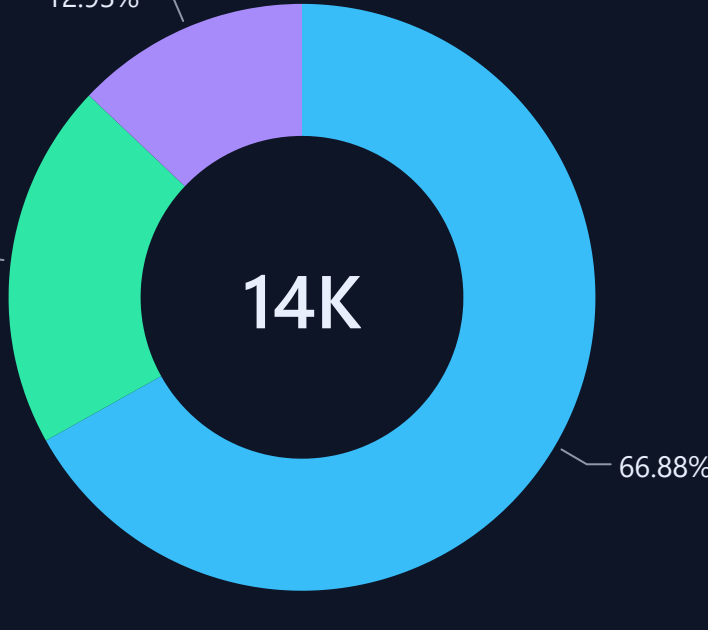
Extra units sold by department. SNAP benefit weeks lift food sales far more than price cuts, at zero cost. Departments without a reliable price effect show no discount bar.

10% discount vs a SNAP week



Where the discounts went

Two thirds of all discount weeks were in food, where shoppers are least responsive to price.



66.9%

Foods

20.2%

Household

12.9%

Hobbies

Department results

Elasticity from a model pooled per department, with a separate sales level per product. Only food shows a reliable price effect, and it is inelastic.

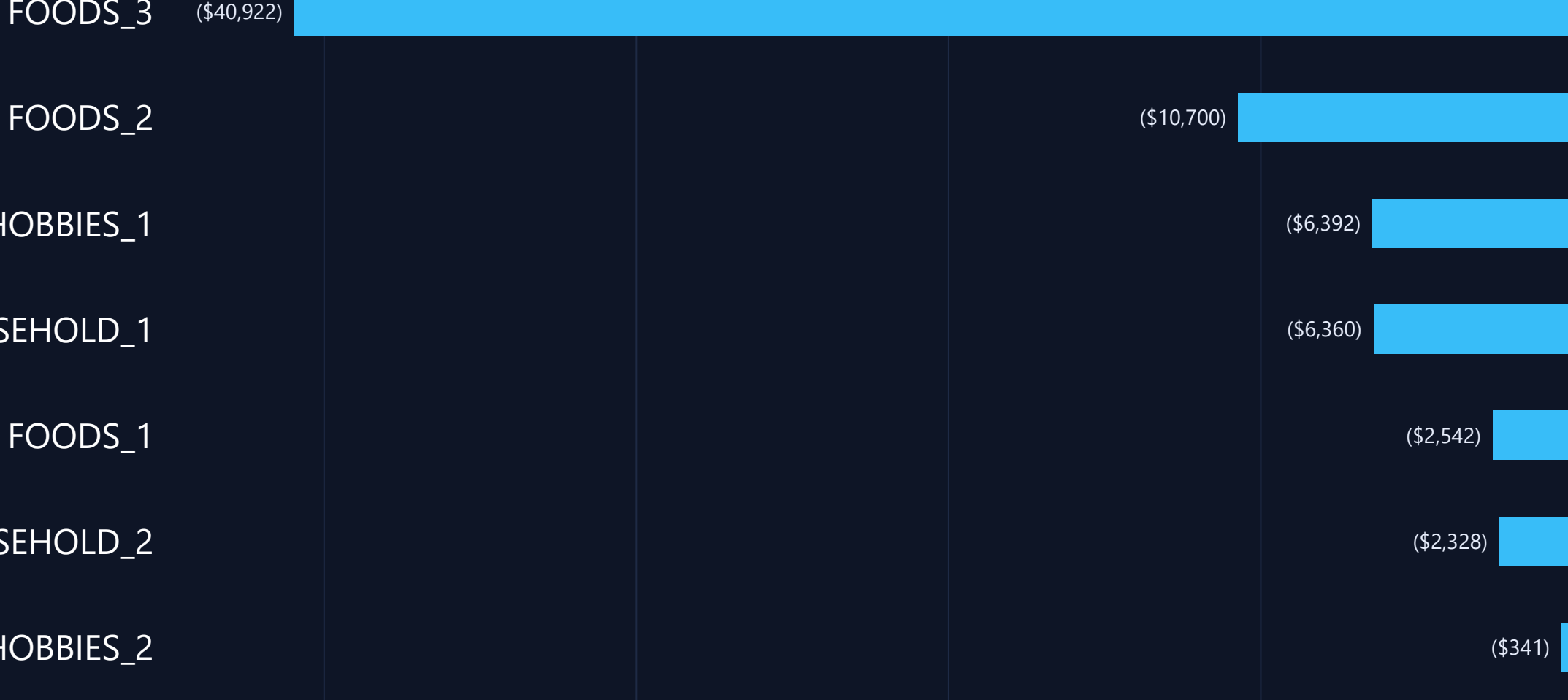
Department	Discount Weeks (count)	Avg discount %	Elasticity	Reliable	Unit Lift from 10% cut (%)	Unit lift in SNAP week (%)
FOODS_1	773	12.7	-0.65	Yes	7.10	0.80
FOODS_3	4806	12.0	-0.59	Yes	6.40	13.50
FOODS_2	3795	9.7	-0.56	Yes	6.10	18.60
HOUSEHOLD_2	1141	11.5	-0.11	No	1.10	2.30
HOBBIES_2	392	20.6	-0.05	No	0.60	1.10
HOUSEHOLD_1	1689	9.3	0.03	No	-0.30	4.70
HOBBIES_1	1420	8.5	0.54	No	-5.50	2.00

Share of total profit lost

25% margin

35% margin

45% margin



Key insights

–\$69.6K

Profit lost to discounts

4–8×

Food shoppers would need to be 4–8× more price-sensitive for a 10% discount to break even.

SNAP beats discounts

+13.5% to +18.6% in FOODS_2 and FOODS_3 (FOODS_1: +0.8%)

Badly timed

26% of discounts in SNAP weeks

Try a food discount

Uses FOODS_3's measured elasticity (–0.59). Move the sliders to see one week of discount versus normal price.

Discount (%)

10

Margin (%)

35

Units sold

+6.4%

Revenue

-4.2%

Gross profit

-24.0%

Break-even elasticity

-3.19

FROM INSIGHT TO ACTION

1. Stop routine food discounts

A 10% food discount adds only 6–7% more units. Ending routine food discounts would have kept about \$54K profit at a 35% margin (2011–2016, one store).

2. Move promotions out of SNAP weeks

Sales in FOODS_2 and FOODS_3 already rise 13.5 to 18.6% in benefit weeks. Save any promotions for the rest of the month.

3. Test before discounting Hobbies and Household

No measurable sales lift was found. Run a small controlled test before continuing; about \$15K profit is at stake, assuming discounts added no extra sales.

The opportunity is not to discount more.

It is to stop paying for sales that would have happened anyway.

Analysis & Dashboard by Vaishnavi Dangeti

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